

Break the Loop with Good Attention: Visual-Attention-Based Probes in GRPO Rewards to Mitigate Repetition Collapse

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Abstract

Reasoning vision–language models trained with GRPO can fail on hard or out-of-distribution inputs by repeating until they exhaust the generation budget instead of answering. We find this collapse coincides with attention drifting from the image to generated text, and that layer–head attention predicts response correctness. Because this signal is imperfect, direct reward shaping can favor confident misreads. We introduce **FT-GRPO**, which uses a frozen attention probe only within task-reward ties through an order-preserving advantage perturbation. Non-tied rollouts remain vanilla-equivalent, and the probe requires no new grounding annotations. On Qwen3-VL-8B-Thinking, FT-GRPO improves in-domain accuracy by 3.4 points and the seven-benchmark OOD mean by 2.4 points, while reducing 4-gram repetition and truncation by roughly one third. A length control costs 20.9 in-domain points, and random or repetition controls do not reproduce the joint gain.

1 Introduction

Group Relative Policy Optimization (GRPO) (Shao et al., 2024) post-trains reasoning models without a separate value model. On difficult or out-of-distribution vision–language inputs, however, models can repeat their reasoning until they exhaust the generation budget instead of answering (Pipis et al., 2025; Xu et al., 2022). We refer to this as *repetition collapse*. It coincides with attention drifting from the image toward generated text (Huang et al., 2026; Chu et al., 2025). Length and repetition penalties risk suppressing useful reasoning; adaptive variants add gates and thresholds but still target surface behavior (Liu et al., 2025; Yu et al., 2025). Visual attention is more mechanism-aligned but fallible, so we need a way to use it without letting its errors override correctness.

We test layer–head visual attention using eight rollouts for each of 6,000 VLAA-Thinking sam-

ples (Chen et al., 2025). A logistic probe predicts content correctness with 0.796 held-out AUROC, while 95.4% of response groups contain a task-reward tie and 34.4% are fully tied, yielding no vanilla-GRPO gradient for those prompts. Probe scores also correlate negatively with repetition and truncation. Because a visually attentive response can still misread the image, the probe measures grounded commitment rather than serving as a correctness oracle. This combination motivates using it only where the verifier is indifferent.

We introduce **FT-GRPO**, which uses the frozen probe only within task-reward ties. A bounded, zero-sum perturbation ranks partial ties without crossing reward levels; for fully tied groups, the probe recovers a scaled signal where vanilla GRPO provides none. Task correctness remains the primary ordering.

On Qwen3-VL-8B-Thinking, FT-GRPO improves held-out in-domain accuracy from 69.3% to 72.7% and the seven-benchmark OOD mean from 47.5% to 49.9%. Gains concentrate on in-domain math (+8.4), multiple choice (+4.4), and OOD MathVision (+6.3). Truncation falls from 18.3% to 12.6% and 4-gram repetition from 15.3% to 10.6%. A length tie-break costs 20.9 in-domain points, while random and repetition controls do not reproduce the joint gain. Together, the controls isolate visual grounding rather than brevity or arbitrary tie perturbation as the useful signal.

2 Preliminary

Let $x = (I, q)$ be an image–question prompt and $y = (y_1, \dots, y_T)$ a response sampled from policy π_θ . We denote the visual- and response-token positions by $\mathcal{V}$ and $\mathcal{R}$, and the normalized causal-attention weight from response position t to context token u at layer ℓ and head h by $a_{t,u}^{\ell,h}$.

For each prompt, GRPO (Shao et al., 2024) samples G responses and assigns reward $r_i = s_i + \eta m_i$,

where s_i is the task-verifier score, m_i indicates valid answer formatting, and $\eta = 0.1$. The verifier is binary for digit, multiple-choice, and math tasks and uses intersection-over-union for localization. Vanilla GRPO standardizes rewards within each group:

$$A_i^{(0)} = \frac{r_i - \bar{r}}{\sigma_r + \epsilon}, \quad (1)$$

where $\bar{r}$ and σ_r are the group mean and unbiased standard deviation. The same advantage is applied to every response token. Tied rewards receive identical advantages; if the whole group is tied, every advantage is zero and the prompt supplies no policy-gradient signal.

3 Method

3.1 Preliminary Experiment: Attention Predicts Correctness

Before RL training, we ask whether visual attention distinguishes responses that coarse outcome rewards treat alike. Frozen Qwen3-VL-8B-Thinking generates eight responses for each of 6,000 VLAA-Thinking questions spanning counting, multiple choice, mathematics, and localization. We split by question into 4,800 discovery and 1,200 held-out questions.

Following Lookback Lens (Chuang et al., 2024), we measure how much each layer and head attends from generated text back to the image:

$$\phi_{\ell,h}(x, y) = \frac{1}{|\mathcal{R}|} \sum_{t \in \mathcal{R}} \sum_{u \in \mathcal{V}} a_{t,u}^{\ell,h}. \quad (2)$$

Qwen3-VL-8B has 36 layers and 32 heads, giving 1,152 features per response. Using format-agnostic correctness labels, we compare mean attention, layer averages, and the full layer-head vector with an ℓ_2 -regularized logistic probe

$$f_\psi(x, y) = \sigma(\mathbf{w}^\top \Phi(x, y) + b). \quad (3)$$

Ground truth is used only to construct discovery labels and is never provided to the probe. We balance correct and incorrect discovery examples within each verifier and cap all verifiers equally, then evaluate on every held-out response.

The full representation reaches 0.796 held-out AUROC, while label shuffling is near chance and averaging loses substantial signal (Table 1). Within the 515 held-out prompts containing both outcomes, the probe ranks a correct response higher in 74.9% of pairs. On vanilla-policy responses, its

Attention features	AUROC
Shuffled labels	0.478
Mean attention	0.592
Layer averages	0.674
Layer-head features	0.796

Table 1: Correctness prediction on held-out responses.

score correlates negatively with 4-gram repetition ($r = -0.47$) and truncation ($r = -0.28$). The probe remains imperfect, and 95.4% of response groups contain a task-reward tie and 34.4% are fully tied. These findings motivate tie-only use; Appendix B.1 gives full diagnostics.

3.2 FT-GRPO: Tie-Constrained GRPO

During policy training, each response is re-forwarded through the frozen base VLM; both the base VLM and probe remain fixed. We first compute the vanilla advantages in Eq. 1. Consider a cluster C of $K \geq 2$ equal advantages in a group containing another reward level. Let Δ_C be the distance to the nearest distinct advantage and let $\rho_i \in \{0, \dots, K-1\}$ be the ascending rank of probe score f_i . We set

$$\tilde{A}_i = A_i^{(0)} + \alpha \Delta_C \left(\frac{\rho_i}{K-1} - \frac{1}{2} \right), \quad i \in C, \quad (4)$$

with $\alpha = 0.5$. The offsets are zero-sum and bounded by $\alpha \Delta_C / 2$: non-tied responses remain unchanged, the group mean is preserved, and for $0 < \alpha \leq 1$ no response can cross a distinct task-reward level. Unavailable scores are excluded; if fewer than two valid scores remain or they are uniform, no perturbation is applied.

When the whole group is tied, $A_i^{(0)} = 0$ and no neighboring reward level defines Δ_C . We instead use

$$\tilde{A}_i = \beta \frac{f_i - \bar{f}}{\max(\sigma_f, \tau)}, \quad \tau = 0.01, \quad (5)$$

where $\bar{f}$ and σ_f are the within-group probe statistics. The scale β is the batch mean of $|A_j^{(0)}|$ over non-tied groups; if the whole batch is tied, $\beta = 0$. Thus the rescue is zero-sum and matches the current advantage scale.

For comparison, our additive baseline uses

$$r_i^{\text{comp}} = m_i[\lambda s_i + (1 - \lambda)f_i] + \eta m_i. \quad (6)$$

where we use $\lambda = 0.7$. Unlike FT-GRPO, it can alter the ordering imposed by task correctness, particularly for continuous verifier scores.

	VLAA-Heldout				Math			General			Medical		
Method	Digit	MCQ	Math	IoU	MathVista	MathVision	MMMU-Pro	MMStar	BLINK	RealWorldQA	VQA-RAD	SLAKE	Overall
Zero-shot (no RL)	94.0	84.0	43.2	24.8	32.8	19.8	40.5	66.1	50.0	54.2	51.1	50.6	50.9
<i>Baseline</i>													
Vanilla GRPO	96.8	87.2	58.8	34.4	33.7	25.4	44.3	68.5	52.9	54.4	53.0	53.4	55.2
Length tie-break	82.4	65.6	9.2	36.4	35.9	37.7	49.0	70.5	52.7	59.7	54.2	52.4	50.5
Random tie-break	96.8	89.6	60.8	34.0	35.2	25.8	44.7	67.0	53.2	55.4	53.2	52.3	55.7
Repetition tie-break	95.2	86.8	57.6	32.8	34.6	24.0	42.2	67.7	51.9	55.4	53.3	52.2	54.5
FT-GRPO (Ours)	94.4	91.6	67.2	37.6	35.5	31.7	48.3	68.6	53.3	57.9	54.0	52.9	57.8

Table 2: Held-out accuracy (%). VLAA is in-domain; other datasets are OOD. Overall averages the 12 columns; best results are bold.

4 Experiment

4.1 Experimental Setup

Training. We fully fine-tune Qwen3-VL-8B-Thinking (Yang et al., 2025) with GRPO in Verl (Sheng et al., 2025) on a stratified VLAA-Thinking subset (Chen et al., 2025) covering four tasks (4,000/400/1,000 train/validation/test). Primary arms train for 250 steps with group size $G = 8$ and share data, rewards, and optimization; only the within-tie signal changes. Details are in Appendix B.

Matched comparisons. The base model receives no RL, while vanilla GRPO uses only task-verifier and format rewards. Three controls reuse FT-GRPO’s tie machinery: *length* favors fewer tokens, *random* uses seeded uniform scores, and *repetition* favors a lower maximum repeated 8-token n-gram frequency. They isolate brevity, arbitrary perturbation, and direct anti-looping; Appendix B.3 gives exact definitions.

Evaluation. We select $S^* = 150$ from vanilla validation only and fix it across arms. Greedy evaluation uses a 4,096-token response budget on VLAA-Heldout and eight OOD benchmarks: MathVista (Lu et al., 2024), MathVision (Wang et al., 2024), MMMU-Pro (Yue et al., 2025), MMStar (Chen et al., 2024), BLINK (Fu et al., 2024), RealWorldQA (xAI, 2024), VQA-RAD (Lau et al., 2018), and SLAKE (Liu et al., 2021). We report task performance, truncation, and 4-gram repetition; OOD means exclude low-collapse SLAKE. Details are in Appendix B.

4.2 Main Results

FT-GRPO improves accuracy in and out of domain. At S^* , FT-GRPO raises in-domain accuracy from 69.3% to 72.7% (+3.4; 95% CI [1.6, 5.4], $p < 0.001$), driven by math (+8.4) and multiple choice (+4.4), while saturated digit

accuracy falls 2.4 points. Mean performance across seven OOD benchmarks rises from 47.5% to 49.9%, including gains of +6.3 on MathVision and +4.0 on MMMU-Pro, while low-collapse SLAKE is nearly unchanged (Table 2).

Generic tie-breaks do not reproduce the joint gain. Length tie-breaking improves the OOD mean to 51.4% but reduces in-domain accuracy to 48.4% (−20.9 points), especially on math. Random and repetition stay near vanilla in domain (70.3% and 68.1% versus 69.3%) and OOD (47.8% and 47.0% versus 47.5%). Among these matched tie-break controls, visual attention is the only signal that improves both axes without suppressing useful reasoning.

FT-GRPO Reduces Collapse without Sacrificing Reasoning. Across seven OOD benchmarks, FT-GRPO reduces 4-gram repetition from 15.3% to 10.6% and truncation from 18.3% to 12.6%, roughly one third (Figure 1). Length reaches 9.0% repetition and 10.9% truncation but incurs the in-domain loss above. Random leaves collapse unchanged, while repetition tie-breaking is slightly worse than vanilla (15.9% repetition; 19.2% truncation).

4.3 Ablation

Tie constraints protect against confident probe errors. The additive reward (Eq. 6) trails FT-GRPO on VLAA-Heldout (70.9% versus 72.7%) but attains the highest Math, General, and Medical OOD averages (39.7%, 61.6%, and 54.3%; Table 3). This stronger aggregate OOD performance comes without an ordering guarantee. On the 21 in-domain prompts where the arms disagree, FT-GRPO is correct on 16 and the additive arm on 5; all 16 additive errors receive high probe scores. Thus additive shaping can favor confident visual misreads, whereas tie constraints preserve the task-reward ordering.

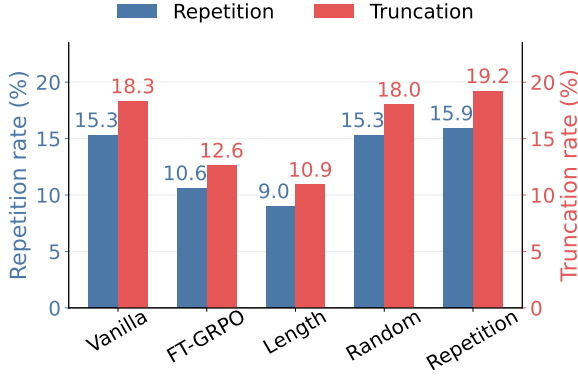


Figure 1: Mean repetition and truncation across the seven OOD collapse benchmarks at S^* ; lower is better.

Method	ID	OOD		
	VLAA	Math	General	Medical
Vanilla GRPO	69.3	34.5	58.6	53.2
<i>Tie-constrained: tie-break strength α</i>				
Faith $\alpha=0.25$	70.0	38.6	59.9	53.6
Faith $\alpha=0.50$ (main)	72.7	38.5	59.9	53.5
Faith $\alpha=0.75$	72.1	39.3	60.4	53.2
Faith $\alpha=1.00$	71.4	39.5	60.0	53.9
<i>Unconstrained reward</i>				
Additive $0.7c+0.3f$	70.9	39.7	61.6	54.3

Table 3: Category-average accuracy (%), grouped as in Table 2. VLAA averages four held-out tasks; Math, General, and Medical average three, three, and two OOD benchmarks. Best values are bold; full results are in Appendix C.

Moderate tie-breaking gives the best trade-off. Every tested strength improves the Math and General averages over vanilla, while Medical performance remains similar or improves (Table 3). The main $\alpha = 0.5$ setting gives the best VLAA-Heldout accuracy (72.7%). Increasing α to 0.75 lowers repetition and truncation from 10.6%/12.6% to 8.8%/10.9% but reduces VLAA accuracy to 72.1%; $\alpha = 1.0$ adds no collapse benefit (Table 5 in Appendix C). We therefore use $\alpha = 0.5$: it maximizes in-domain accuracy while retaining gains across all three OOD groups and substantial collapse reduction.

5 Related Work

Attention and internal signals for RL. Look-back Lens (Chuang et al., 2024) uses attention to diagnose contextual hallucination after generation. Recent RL methods instead exploit internal signals during training: ProFIL (Parekh, 2026) probes attention to suppress post-commitment reasoning; Saliency-R1 (Gong et al., 2026) rewards agree-

ment with annotated regions; Reinforced Attention Learning (Li et al., 2026a) directly optimizes attention; and PAPO (Wang et al., 2026) rewards sensitivity to masking the image. We require no new annotations: a frozen layer-head attention probe is trained from existing verifier labels and influences learning only where the verifier is indifferent. We additionally do not optimize outside of correctness.

Reward ties in GRPO. Rank-based methods recover signal from coarse rewards using preference models or lexicographic correctness and length rules (Zhu et al., 2026; Niu et al., 2026). Fully tied groups are otherwise discarded by methods such as DAPO (Yu et al., 2025) or rescued with virtual samples by AVSPO (He et al., 2026). FT-GRPO instead ranks only task-reward ties with visual attention: partial-tie perturbations cannot reverse the verifier ordering, while the all-tied rescue (Eq. 5) recovers a grounded learning signal.

Repetition collapse. Repetitive decoding arises from self-reinforcing generation (Holtzman et al., 2020; Xu et al., 2022) and reappears as entropy collapse in RL-trained reasoners (Cui et al., 2025). Length and repetition penalties (Li et al., 2026b; Ginart et al., 2025) directly suppress these symptoms but can also remove useful reasoning; entropy and optimization fixes remain content-blind (Cui et al., 2025; Yu et al., 2025). Our method instead favors visually grounded commitment within correctness ties, reducing loops without directly optimizing brevity or repetition.

6 Conclusion

We identify layer-head visual attention as a signal of grounded commitment and introduce FT-GRPO, which uses a frozen probe only within task-reward ties. This constraint preserves correctness ordering while recovering signal from fully tied groups. Case studies show why it matters: attentive responses can still misread an image, while correct responses can emerge after low-attention loops. On Qwen3-VL-8B-Thinking, FT-GRPO improves both in-domain and OOD performance while reducing repetition and truncation by roughly one third; generic length, random, and repetition controls do not match this trade-off. More broadly, tie-constrained advantages offer a controlled way to exploit useful but imperfect internal signals when outcome rewards are coarse.

Limitations

The current experiment tests on one seed, with one model. Previous experiments have held up results on smaller models, but those configurations are not directly comparable to this setup so they are not displayed. The probe is also model-family specific (Qwen-trained probe must be retrained to transfer to Llama family.) Probes cannot be transferred down model sizes within the same model family (8B→4B) but can be transferred within type (Thinking→Instruct).

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A Case Study

We score policy responses with the frozen probe and inspect what it rewards.

A.0.1 Ex. 1

Ex. 1 is a physics MCQ question. We test Vanilla GRPO, Additive, and faith tiebreak arms. In this response, the vanilla-GRPO-trained model misreads the graph and commits 894 token wrong answer.

Prompt: What trend from graphs (c)/(d) as temperature increases? (GT: B)

"Vanilla GRPO (answers D) (faith 0.380):

... ϵ_1 decreases ... as temperature increases, Γ decreases. Wait, but option B says ... the linewidth Γ_1 increases. But if Γ_1 decreases, then B is wrong. Wait, maybe I misread ... Wait, let's check the graphs again ... both decrease. So option D.

To the probe, misreading and looping are attention signals indicative of an incorrect response. The same question answered by tiebreak is shorter (229 tokens) and reads the chart correctly, then commits.

Faith tie-break (answers B) (faith 0.750):

... as temperature increases, the ϵ_1 data points ... decrease ... so anisotropy ϵ_1 decreases. ... graph (d): ... as temperature increases, Γ increases ... So the trend is ϵ_1 decreases and Γ increases. That matches option B.

When the Additive reward arm (Eq. 6) responds to the same question, it shows this reward structure lacks the guarantee and can rank a confident misread highly. This answer is the shortest of all, however. (105 tokens)

Additive arm (answers D) (faith 0.753):

... graph (c) shows ϵ_1 (anisotropy) vs temperature; as temperature increases, the data points decrease. Graph (d) shows Γ_1 (linewidth) vs temperature; as temperature increases, the data points also decrease. So both ϵ_1 and Γ_1 decrease with temperature. That matches option D.

Placing the probe score directly in the reward teaches the additive arm to commit more concisely than tie-break (229 vs 105 tokens, faith 0.75 vs 0.753), but with no guarantee that this commitment respects correctness.

B Implementation and Metric Details

B.1 Preliminary Probe Experiment

The frozen base model generates one rollout at temperature 0.7, five at 1.0, and two at 1.3 per question. The prompt-level split contains 38,400 discovery and 9,600 held-out responses. Labels ignore answer formatting: digit, multiple-choice, and math responses use content correctness, while localization uses IoU ≥ 0.5 . We fit logistic regression with $C = 1$ using 474 correct and 474 incorrect responses from each task. The probe satisfies three validation checks: held-out AUROC above 0.55, mean within-prompt score standard deviation above 0.05, and shuffled-label AUROC below 0.55.

Among the 515 held-out prompts containing both correct and incorrect responses, the probe ranks a correct response higher in 74.9% of pairs. On 1,000 vanilla-policy responses, its score correlates with correctness ($r = 0.49$) and negatively with 4-gram repetition ($r = -0.47$), response length ($r = -0.45$), and truncation ($r = -0.28$); all correlations have $p < 0.001$. These diagnostics indicate grounded commitment rather than a pure length or repetition heuristic, but do not make the probe a correctness oracle.

B.2 Training

We use the 4,000/400/1,000 train/validation/test split of the stratified VLAA-Thinking subset and implement full-parameter GRPO with Verl on two H200 GPUs. Every variant uses seed 42 and trains for 250 steps with a batch of 16 prompts, eight responses per prompt, and a PPO mini-batch of 16. We use a learning rate of 10^{-6} with cosine decay and 5% warmup, KL coefficient 0.04, entropy coefficient 0.001, and maximum prompt and response lengths of 4,096 within an 8,192-token envelope. The required format is `<think>...</think><answer>...</answer>`. Data order, optimization, token budgets, and checkpoint cadence are shared across arms.

B.3 Control Tie-Breakers

Each control substitutes its score for the probe output while retaining the same tie-breaking estimator. Length uses negative response-token count. Random uses seeded independent uniform scores. Repetition uses the negative maximum repeated 8-token n-gram frequency: for a response with at least 16 tokens, it divides the largest count of any

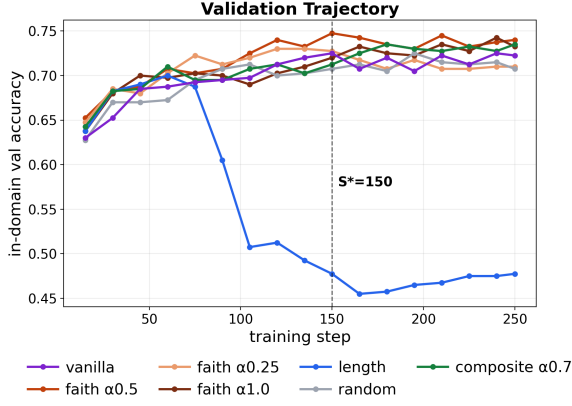


Figure 2: Validation trajectories over 250 training steps; $S^* = 150$ is selected from vanilla GRPO only.

token 8-gram by the number of 8-grams; shorter responses receive zero. This training signal targets verbatim loops. It is distinct from the broader 4-gram repetition evaluation metric defined below.

B.4 Checkpoint Selection

Using only vanilla-GRPO validation, we compute a three-checkpoint moving average over the four in-domain tasks and select the first checkpoint after the last relative improvement above 1%. This chooses $S^* = 150$, which is fixed for every arm.

B.5 Evaluation and Answer Scoring

We decode greedily with temperature 0, $\text{top}_p=1.0$, a maximum prompt length of 4,096, and a maximum response length of 4,096. Scoring is strict and identical across variants. We identify closed `<answer>`. . . `</answer>` spans case-insensitively and score only the final closed span; a response without one receives zero credit. Digit answers use exact match, and multiple-choice answers use the first standalone option letter from A–E in the final answer span. Mathematical answers are scored by symbolic equivalence with `math_verify`, including equivalent fractional, decimal, and boxed forms. Localization uses intersection-over-union, with $\text{IoU} \geq 0.5$ treated as correct when a binary decision is required. For short-answer and medical benchmarks, we use exact match for yes/no and integer targets, entity-name containment for single-token targets, and token-level F1 otherwise. The in-domain confidence interval uses paired bootstrap resampling over prompts; significance is assessed with McNemar’s test.

B.6 Collapse Metrics

We report two response-level measures of repetition collapse. A response is counted as truncated when it does not close its reasoning block with `</think>` before reaching the generation budget. For repetition, we use 4-gram repetition (Welleck et al., 2020). Let $x = (x_1, \dots, x_T)$ be the whitespace-tokenized response and $\mathcal{G}_n(x)$ the multiset of its contiguous n -grams. We define

$$\text{REP-}n(x) = 1 - \frac{|\text{set}(\mathcal{G}_n(x))|}{|\mathcal{G}_n(x)|}, \quad (7)$$

and report $n = 4$. The score is zero when no 4-gram repeats and approaches one as repetition increases; responses shorter than four tokens receive zero.

C Full Ablation Results

D Probe scores over Baseline Responses

The following responses illustrate edge cases for the probe (Ex. 2 – 4) and are made by the vanilla GRPO arm.

D.0.1 Ex. 2

A long math rollout the probe scores highly (988 tokens).

Prompt: Solve for $\cos^2 O$.

GT: $\cos^2 O = \frac{2809}{4356}$ (faith 0.835):

"... by the Law of Cosines, $19^2 = 11^2 + 9^2 - 2 \cdot 11 \cdot 9 \cos O$, so $361 = 202 - 198 \cos O$ and $\cos O = -\frac{159}{198} = -\frac{53}{66}$.

Wait, let me verify the Law of Cosines step again to make sure I didn’t make a mistake: $361 = 121 + 81 - 198 \cos O$, so $361 - 202 = -198 \cos O$, giving $\cos O = -\frac{53}{66}$. Correct. Then $\cos^2 O = \left(\frac{53}{66}\right)^2 = \frac{2809}{4356}$."

The probe does not penalize this response for length because it finishes the question without repeatedly attending to its own previous text. Even when the model self-doubts, each step builds off of the last.

D.0.2 Ex. 3

An incorrect response that was scored highly, using 349 tokens.

Prompt: Find the area of the parallelogram.

GT: $C = 1520$ (Faith 0.839)

Method	VLAA-Heldout					Math			General			Medical	
	Avg.	Digit	MCQ	Math	IoU	MathVista	MathVision	MMMU-Pro	MMStar	BLINK	RealWorldQA	VQA-RAD	SLAKE
Vanilla GRPO	69.3	96.8	87.2	58.8	34.4	33.7	25.4	44.3	68.5	52.9	54.4	53.0	53.4
Faith $\alpha=0.25$	70.0	87.6	86.4	66.8	39.2	35.7	31.6	48.5	68.7	52.6	58.4	55.3	51.8
Faith $\alpha=0.50$ (main)	72.7	94.4	91.6	67.2	37.6	35.5	31.7	48.3	68.6	53.3	57.9	54.0	52.9
Faith $\alpha=0.75$	72.1	96.4	88.8	65.6	37.6	36.1	33.9	48.0	69.1	52.8	59.3	53.9	52.5
Faith $\alpha=1.00$	71.4	96.4	88.4	66.0	34.8	36.4	33.8	48.3	68.9	52.9	58.3	54.7	53.1
Additive $0.7c+0.3f$	70.9	94.4	91.2	63.2	34.8	35.4	36.1	47.7	69.7	53.1	62.1	56.1	52.5

Table 4: Per-benchmark accuracies (%) underlying Table 3. Best values in each column are bold.

Method	4-gram rep. ↓	Trunc. ↓
Vanilla GRPO	15.3	18.3
Faith $\alpha=0.25$	10.5	12.4
Faith $\alpha=0.50$ (main)	10.6	12.6
Faith $\alpha=0.75$	8.8	10.9
Faith $\alpha=1.00$	9.3	11.2
Additive $0.7c+0.3f$	7.7	10.4

scored against other correct responses, and when there are none, it gets the highest reward.

Table 5: Mean collapse rates (%) across the seven OOD collapse benchmarks for the ablation arms.

... the area of a parallelogram is base times height... the base is probably the 38 m side, and the height is the 28 m... area = base $\times$ height = 38×28 . 38 times 28. 30 $\times$ 28 is 840, 8 $\times$ 28 is 224, so $840+224 = 1064$. So the area is 1064... the answer should be A.

While the final answer was wrong in this case because the model incorrectly assumed the height of the parallelogram to be 28m, this response works through each step with clean logic, never falling into a loop. We re-rank within ties with faith so that within wrong responses, the policy can be encouraged towards those that perceive and respond over those that loop endlessly.

D.0.3 Ex. 4

This MCQ response was correct (it answered D after 2,738 tokens of looping) but the answer had a lower faith score of 0.357.

Prompt: What trend in time over successive iterations in figures A and B?

GT: $D = \text{Time decreases}$. (Faith 0.357)

... **Wait**, but the question is about the trend... **Wait**, maybe I misread... **Wait**, none of the options fit... **Wait**, maybe the question is wrong, but let's check again...

The continuous back-and-forth makes this response less preferable to other correct responses that are more concise. In practice, this response is only